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EXPERIMENT: 08

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1. Aim:-

To design a high-level architecture for a scalable stock trading platform (similar to Zerodha, Groww, and Upstox) that supports real-time stock trading, order management, and portfolio tracking with high availability and low latency.

2. Objectives:-

Theobjectivesof this experiment are:

1. To understand the architecture of a real-time stock trading system.
2. To design APIs for user authentication, KYC verification, market data, and order execution.
3. To analyze functional and non-functional requirements of a trading platform.
4. To study how real-time data streaming and event-driven systems work.
5. To ensure consistency in financial transactions while maintaining scalability.
6. To understand how portfolio and trade management systems operate.

3. Procedure:-

• Studied real-world stock trading platforms such as Zerodha, Groww, and Upstox to understand system behavior.
• Identified key system entities including Users, Stocks, Orders, Trades, Portfolio, and Watchlist.
• Analyzed the complete workflow from user registration → KYC → viewing stocks → placing orders → trade execution.
• Gathered functional requirements such as real-time stock updates, order placement, and portfolio tracking.
• Analyzed non-functional requirements including low latency, high availability, fault tolerance, and scalability.
• Designed RESTful APIs for user management, market data, portfolio, and watchlist features.
• Studied WebSocket-based real-time updates for stock prices.
• Understood event-driven architecture using message queues (Kafka) for trade execution and notifications.
• Evaluated trade-offs between strong consistency (orders, trades) and eventual consistency (analytics, notifications).

4. Functional Requirements:-

Users should be able to register,log in, and complete KYC verification.
•Users should be able to view real-time stock prices and historical data.
•The system should support placing, modifying, and canceling orders.
•Orders should provide accurate real-time status updates.
•Users should be able to maintain a watchlist with live updates.
•The system should provide a dashboard to track holdings, trade history, and P&L.
•Portfolio management should reflect executed trades instantly.

5. Non Functional Requirements:-

•The system must support millions of users simultaneously.
•Real-time stock updates should have very low latency (<100 ms).
•High availability (99.9%+) must be ensured for trading operations.
•Strong consistency is required for order execution and trades.
•The system should be fault tolerant and resilient to failures.
•Scalability should be achieved using microservices architecture.
•Secure communication and authentication (JWT, HTTPS) must be implemented.

6. Core Entities:-

•User
•Stock
•Order
•Trade
•Portfolio
•Watchlist

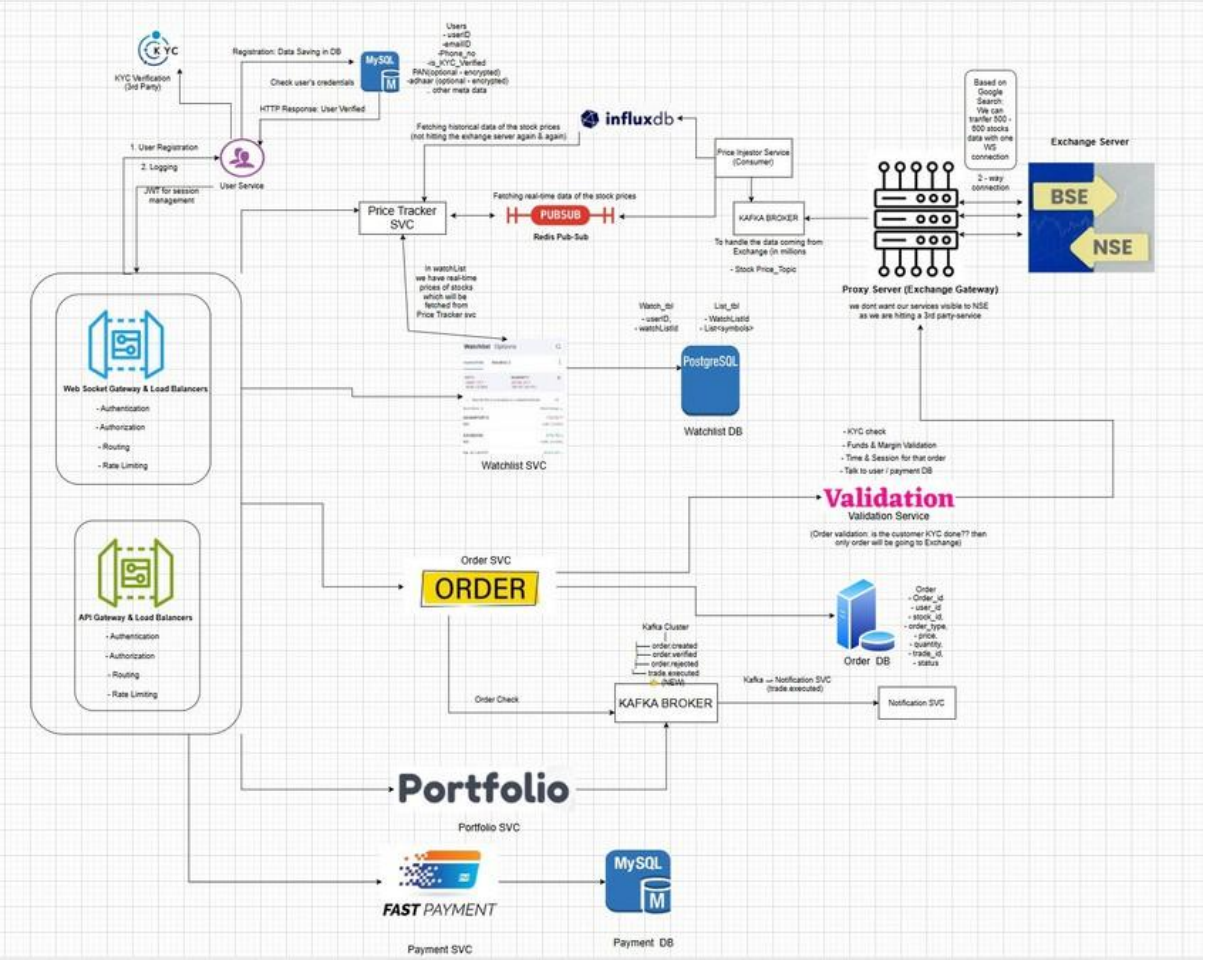
7. API Design:-

User APIs:
•POST /auth/signup
•POST /auth/login
•POST /auth/verify-KYC
Market Data APIs:
•WS /api/v1/stocks → Real-time stock updates
•GET /api/v1/stocks → Get all stocks
•GET /api/v1/stocks/{stock_symbol_id} → Stock details
•GET /api/v1/stocks/{stock_symbol_id}/history → Historical data
Order APIs:
•POST /api/v1/orders → Place order
•PUT /api/v1/orders/{order_id} → Modify order
•DELETE /api/v1/orders/{order_id} → Cancel order
•GET /api/v1/orders → Get user orders

Portfolio APIs:

•GET /api/v1/portfolio
•GET /api/v1/portfolio/holdings
•GET /api/v1/portfolio/positions
•GET /api/v1/portfolio/performance (Optional)

Watchlist APIs:

•GET /api/v1/watchlists
•POST /api/v1/watchlists/{watchlist_id}/symbols
•DELETE /api/v1/watchlists/{watchlist_id}/symbols/{symbol}

8. HLD:-